AutoRef: Harness Optimization for Agentic Multi-Reference Image Generation

Executable-code search · frozen generator + reasoner · train feedback ≠ validation selection

Reel · 65 / 100

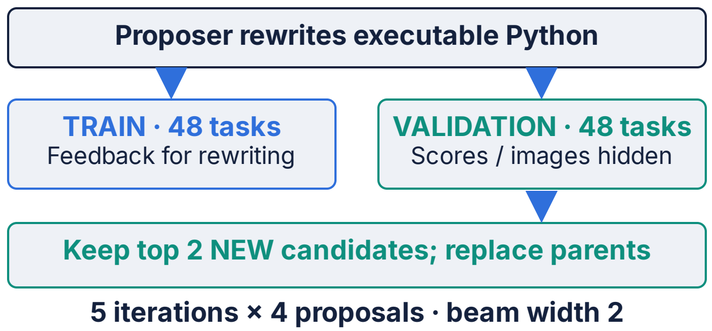
Retained artifact

Paper · v1

Code

1 · Search the harness, freeze the models

Change prompts, references, selection and control flow—not weights.



Training history includes rejects. Validation returns survivor code, not scores. Final pick: top validation member of final beam.

Retained runtime · three generated images

Ground references → drafts A + B
Scene view + background/style canvas.

↓

Winner → up to five complaints → draft C

↓

Select final image: fewer hard failures; tied challenger must win both display orders.

Fallbacks: resample A if no canvas reference; resample winner if no complaints. Reasoner: GPT-5.5. Evaluator: Qwen3-VL-8B.

2 · Held-out gain beyond more samples

5.72 → 7.37

FLUX.2 [klein] 4B · MultiBanana
133 test tasks · four references

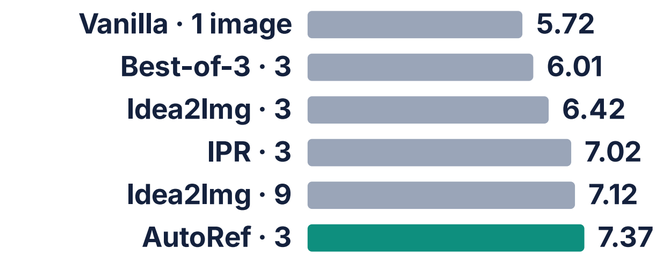


Table 6 · equal mean over four task types. Same generator / reasoner. Image counts ≠ total compute.

Retained-program ablations · Tables 4 / 9



Three-image ablations replace removed steps with resampling. Grounding-only is not a pure leave-one-out.

Search controls: Meta-Harness 6.23 · greedy 6.67.
Same budget stated; total compute not measured.

3 · Scope, controls and failure boundaries

Broad on the main split, not every column

All four 4B task types improve (Table 1). But local **7.87** trails GPT-Image-1.5 **8.12**; global **7.64** trails Seedream 4.5 **7.83**.

Transfer is real—and uneven

Unchanged harness: 9B **5.78→7.10**; Qwen **4.44→5.69** on four references.

> Five-reference Qwen: **1.89→1.87**.
> All three drafts hard-fail on **87/96** tasks.

OmniContext: 4B **8.30→8.85**, but single-character **9.22→8.94**; scene-object **7.47→8.89**. Different evaluator, unseen benchmark.

What baseline matching actually means

Baselines see references: Best-of-3 selector, GEMS verifier, IPR planner, Idea2Img feedback. IPR is reimplemented; matched Idea2Img uses one round, no feedback.

Meta-Harness sees same-task feedback/ranking; greedy keeps the split but one parent. Meta uses five iterations here. Separate baseline tuning budgets are unreported.

Small differences need uncertainty

Four raters, 50 tasks per comparison: vs Nano Banana Pro, **46% win / 40% loss**. No CIs or per-cell seed/run counts reported.

Tables 1–3, 6–11; §6.6; Apps G–I. Qualitative examples favor large gaps.

Takeaway · improve the retained program, not the improver

Split proposal feedback from survival selection. The retained three-image harness beats simple resampling and matched-image controls—but the headline also buys reasoning and more generation.

AutoRef · arXiv 2609.35530v1 · Oshima, Onoda, Iwasawa, Suzuki, Matsuo, Furuta · Original diagrams; no paper figures reused.

Our read · 65 / 100 · ★★ · retained

Reports: held-out tests, runtime controls, component ablations, transfer and human ratings.

Does not report: search variance, total optimization cost or content-overlap audit. Validation is selection evidence.